

Лабораторийн ажил №1

МТЭС, МКУТ, Программ хангамж

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Ажлын зорилго

Линукс (Ubuntu) үйлдлийн системийн орчинд сүлжээний тохиргоог харуулах командуудтай танилцах, Packet Internet Groper (ping) команд, Traceroute (tracert) команд, nslookup командуудыг турших, Packet tracer программтай танилцах, Wireshark программтай танилцах.

Үндсэн ойлголт

1. Ping команд нь IP сүлжээ дээр хост хооронд холболтыг шалгах, мөн сүлжээнд илгээгч төхөөрөмжөөс хүлээн авагч төхөөрөмж хооронд мессэж дамжуулах үеийн Round-trip time (RTT) -ийг хэмжихэд ашигладаг. ping команд нь Internet Control Message Protocol (ICMP) протоколыг ашиглах бөгөөд хүлээн авагч төхөөрөмж рүү ICMP echo request пакетыг илгээж, хариу ICMP echo reply пакетыг хүлээн авна.
2. Traceroute нь компьютерээс тодорхой сервер, вэбсайт руу мэдээлэл очихдоо ямар замаар (ямар рутерүүдээр дамжин) хүрч байгааг шат дараатай харуулдаг команд юм.
3. nslookup гэдэг нь *Name Server Lookup* гэсэн үгийн товчлол бөгөөд DNS (Domain Name System)–ээс мэдээлэл авахад ашиглагддаг команд юм. DNS нь вебсайтын нэрийг (жишээ нь: google.com) IP хаяг руу хөрвүүлдэг систем. Nslookup нь энэ хөрвүүлэлтийг шалгах, туршихад хэрэглэгддэг.

Туршилт

1. #ping [option] target_name (google.com, sisi.num.edu.mn, г.м.)

Ping командын ашиг дараах options-уудыг туршиж үзэв.

Options	Option -ний тайлбар	Туршилтын үр дүн
-n	DNS хөрвүүлэхгүй зөвхөн IP address харуулна.	<pre> munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx: \$ ping google.com PING google.com (142.250.198.142) 56(84) bytes of data: 64 bytes from nchkgb-aj-in-f14.1e100.net (142.250.198.142): icmp_seq=1 ttl=112 t ime=119 ms 64 bytes from nchkgb-aj-in-f14.1e100.net (142.250.198.142): icmp_seq=2 ttl=112 t ime=234 ms 64 bytes from nchkgb-aj-in-f14.1e100.net (142.250.198.142): icmp_seq=3 ttl=112 t ime=135 ms ^C --- google.com ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 6241ms rtt min/avg/max/ndev = 119.281/162.772/233.538/50.474 ms munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx: \$ ping google.com PING google.com (142.250.198.142) 56(84) bytes of data: 64 bytes from nchkgb-aj-in-f14.1e100.net (142.250.198.142): icmp_seq=1 ttl=113 t ime=281 ms 64 bytes from nchkgb-aj-in-f14.1e100.net (142.250.198.142): icmp_seq=2 ttl=113 t ime=97.2 ms 64 bytes from nchkgb-aj-in-f14.1e100.net (142.250.198.142): icmp_seq=3 ttl=113 t ime=355 ms ^C --- google.com ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 2003ms rtt min/avg/max/ndev = 97.213/217.577/354.917/105.889 ms </pre>
-t	Ping google.com команд нь тухайн нэг хост руу тасралтгүй ping илгээж хариу авна. Ctrl + C хүсэлтийг зогсооно.	

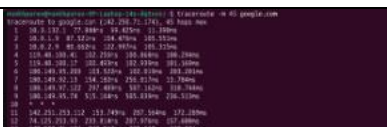
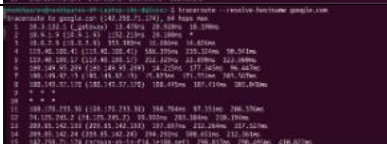
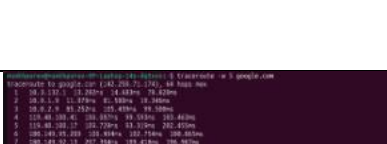
-i	Ping -i 2 google.com Коммандаар пакет илгээх хоорондын зайг нь тохируулна. -i нь секундыг илэрхийлнэ. Сэкунд нь 0.2 оос багагүй утга байна.	<pre>munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ ping -i 2 google.com PING google.com (142.250.198.142) 56(84) bytes of data. 64 bytes from nchkgb-aj-lin-f14.1e100.net (142.250.198.142): icmp_seq=1 ttl=112 tme=145 ms 64 bytes from nchkgb-aj-lin-f14.1e100.net (142.250.198.142): icmp_seq=2 ttl=112 tme=190 ms 64 bytes from nchkgb-aj-lin-f14.1e100.net (142.250.198.142): icmp_seq=3 ttl=112 tme=236 ms ^C --- google.com ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 4005ms rtt min/avg/max/ndev = 144.648/190.459/236.477/37.489 ms munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ ping -l 3 google.com PING google.com (142.250.71.206) 56(84) bytes of data. 64 bytes from nchkgb-aa-lin-f14.1e100.net (142.250.71.206): icmp_seq=3 ttl=116 tme=296 ms 64 bytes from nchkgb-aa-lin-f14.1e100.net (142.250.71.206): icmp_seq=2 ttl=116 tme=296 ms 64 bytes from nchkgb-aa-lin-f14.1e100.net (142.250.71.206): icmp_seq=1 ttl=116 tme=296 ms</pre>
-l, -f	suppress loopback of multicast packets, flood ping	<pre>munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ sudo ping -f google.com [sudo] password for munkhpurev: PING google.com (142.250.71.206) 56(84) bytes of data.^C --- google.com ping statistics --- 2220 packets transmitted, 2204 received, 0.720721% packet loss, time 28260ms rtt min/avg/max/ndev = 69.817/98.287/179.758/14.944 ms, pipe 15, ipg/ewm 12.735 /93.887 ms</pre>
-w	reply wait <deadline> in seconds	<pre>munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ ping -w 3 google.com PING google.com (142.250.71.206) 56(84) bytes of data. 64 bytes from nchkgb-aa-lin-f14.1e100.net (142.250.71.206): icmp_seq=1 ttl=116 tme=280 ms 64 bytes from nchkgb-aa-lin-f14.1e100.net (142.250.71.206): icmp_seq=2 ttl=116 tme=83.6 ms 64 bytes from nchkgb-aa-lin-f14.1e100.net (142.250.71.206): icmp_seq=3 ttl=116 tme=330 ms ^C --- google.com ping statistics --- 3 packets transmitted, 3 received, 0% packet loss, time 2004ms rtt min/avg/max/ndev = 83.579/231.119/329.925/106.310 ms</pre>

Ping command options

Талбарууд	Төхөөрөмжийн тохиргоо	Тайлбар (хэдэн битийн урттай, үүрэг зэргийн бичнэ.)
IPv4 хаяг	Ip addr show inet 10.3.201.213/22	32 битийн хаягтын форматтай
Subnet mask	Ip addr show 23 = 255.255.254.0	the leftmost 24 bits of the 32-bit quantity define the subnet address.
MAC address	Ip link show link/ether 50:5a:65:cc:65:a1	
Default gateway	ip route grep default default via 10.3.200.1	
Host Name	Munkhpurev	
Linkn local IPv6 хаяг	Ip addr show inet6 fe80::a413:9cee:8458:826f/64	128 битийн хаягтын форматтай
DNS server	Resolvectl Current DNS Server: 10.0.80.8	
DHCP server	ip route grep default dhcp src 10.3.201.213	

Төхөөрөмжийн сүлжээний тохиргоо

options	tailbar	Turshiltiin ur dun
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-h	Used to specify the max number of hops to search for the target default is 64	
-d	this option instructs the tool not to perform DNS lookups on each IP address, thereby displaying only the IP addresses of the routers along the path	
-w	Wait timeout seconds for responses (default is 3).	

Tracert командын options

Домэйн нэр	IP хаягууд
Sisi.num.edu.mn	<pre>munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ nslookup sisi.num.edu.mn Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: sisi.num.edu.mn canonical name = sisi.num.edu.mn.86961.fortwebcloud.net. sisi.num.edu.mn.86961.fortwebcloud.net canonical name = lb-2-ap-east-1.aws.waas-online.net. Name: lb-2-ap-east-1.aws.waas-online.net Address: 16.162.201.188</pre>
Google.com	<pre>munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ nslookup google.com Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: Name: google.com Address: 142.250.198.46 Name: google.com Address: 2404:6800:4005:826::200e</pre>
Youtube.com	<pre>munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~\$ nslookup youtube.com Server: 127.0.0.53 Address: 127.0.0.53#53 Non-authoritative answer: Name: youtube.com Address: 142.250.199.206 Name: youtube.com Address: 2404:6800:4005:81b::200e</pre>

Nslookup

```
munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~$ nslookup google.com
Server:      127.0.0.53
Address:     127.0.0.53#53

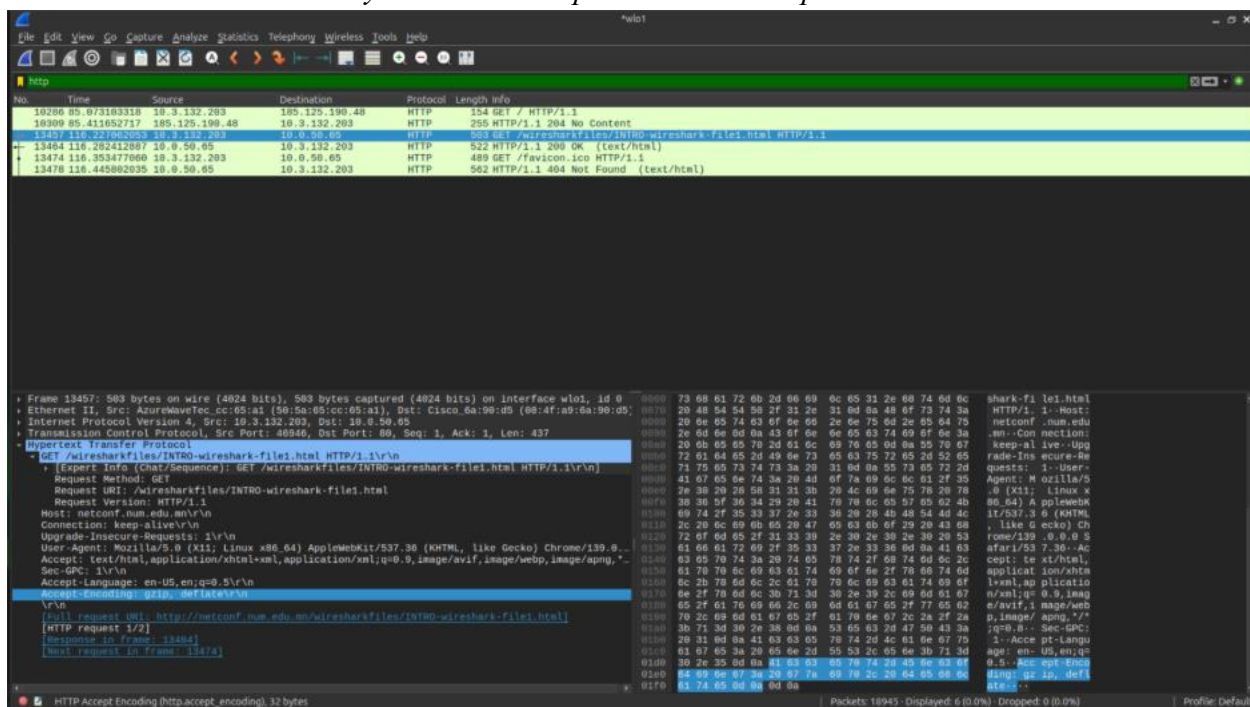
Non-authoritative answer:
Name:   google.com
Address: 142.250.198.46
Name:   google.com
Address: 2404:6800:4005:826::200e
```

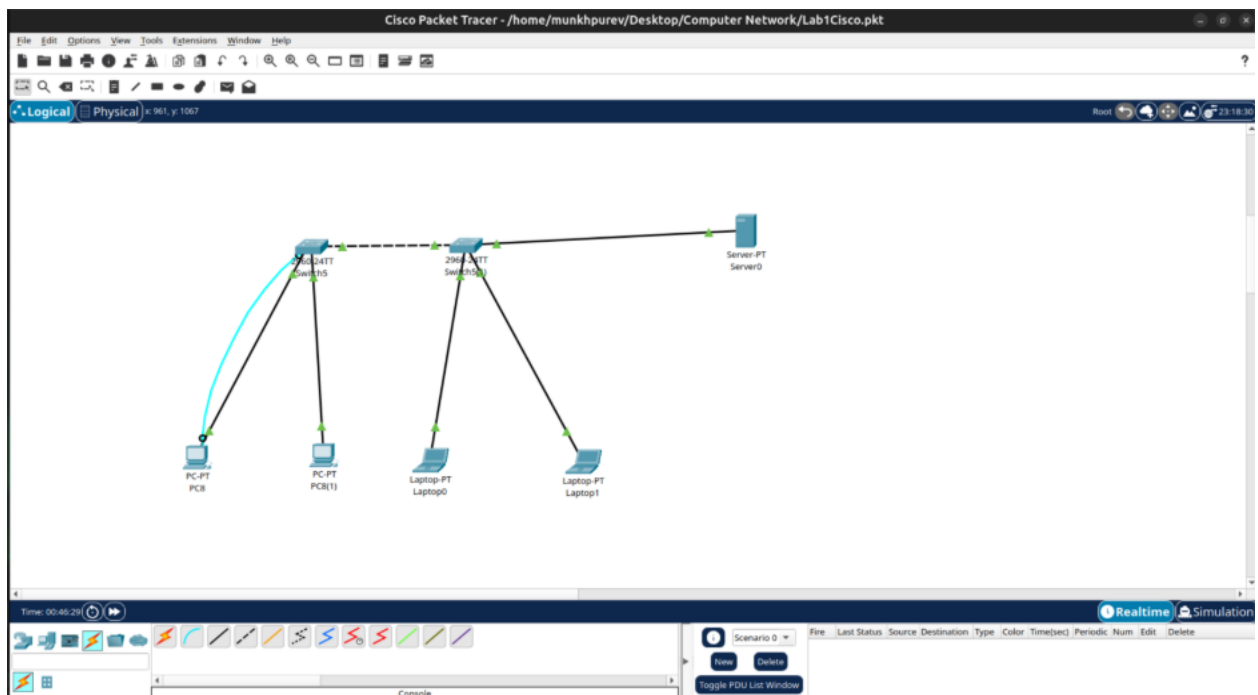
```
munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~$  
munkhpurev@munkhpurev-HP-Laptop-14s-dq5xxx:~$ traceroute cisco.com  
traceroute to cisco.com (72.163.4.185), 64 hops max  
 1  172.20.10.1  5.677ms  4.646ms  4.266ms  
 2  *  10.150.0.254  462.469ms  476.297ms  
 3  10.150.0.2  240.483ms  45.506ms  67.177ms  
 4  103.170.17.45  199.253ms  96.532ms  313.057ms  
 5  180.149.95.33  189.405ms  111.844ms  312.857ms  
 6  180.149.92.9  403.596ms  312.211ms  267.135ms  
 7  180.149.95.246  450.698ms  301.212ms  311.838ms  
 8  62.115.181.126  404.676ms  283.323ms  335.941ms  
 9  * * *  
10  4.69.208.229  369.978ms  312.779ms  404.088ms  
11  4.59.34.66  312.464ms  403.942ms  307.821ms  
12  128.107.2.5  516.416ms  404.185ms  517.272ms  
13  72.163.0.102  409.733ms  613.911ms  404.545ms  
14  72.163.0.186  373.340ms  450.847ms  506.517ms  
15  72.163.2.202  414.273ms  72.163.3.2  404.738ms  312.327ms  
16  * * *  
17  * * *  
18  * * *  
19  * * *  
20  * ^C
```

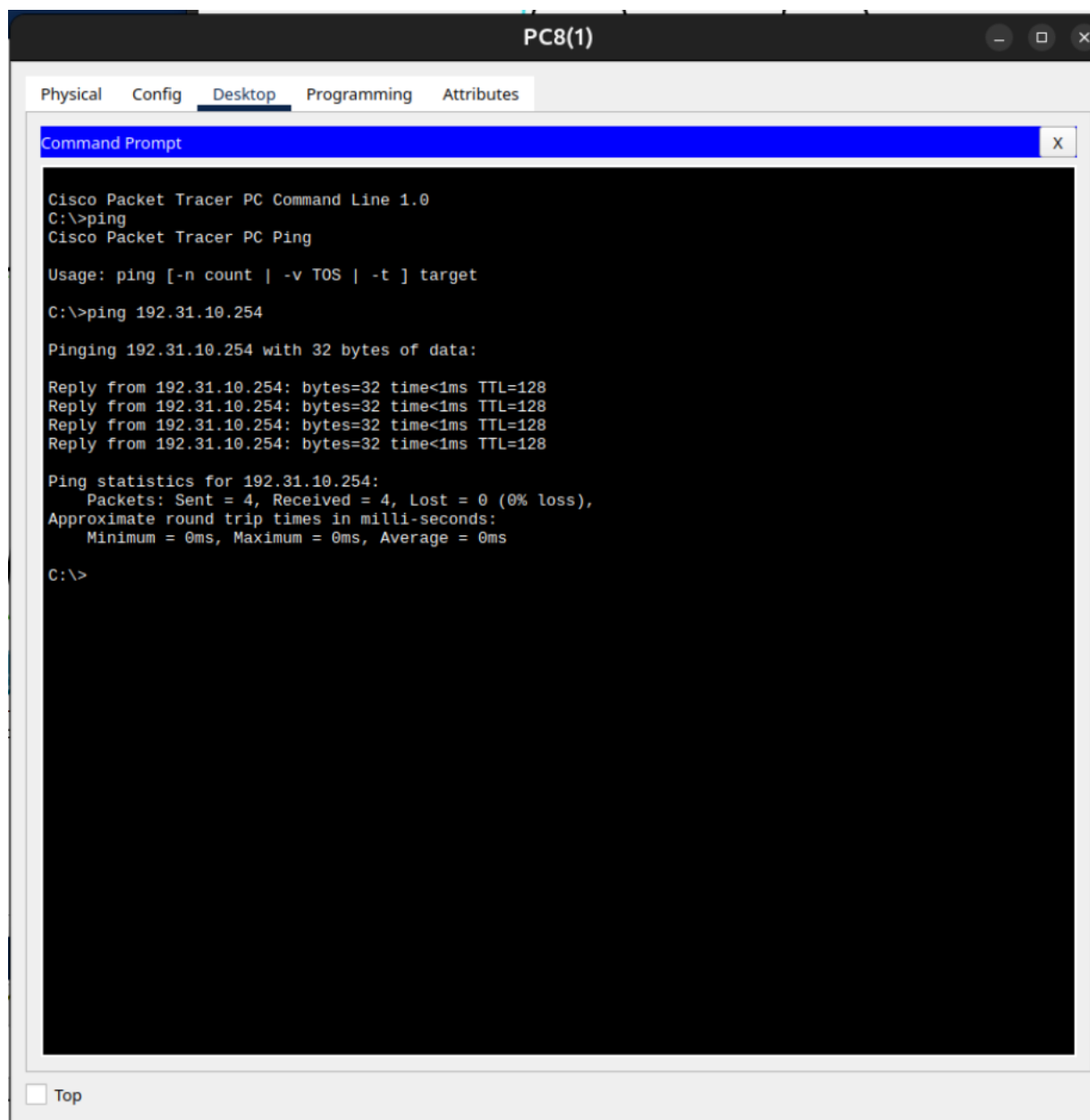
```
traceroute to gaia.cs.umass.edu (128.119.245.12), 64 hops max
 1  172.20.10.1  8.593ms  4.218ms  4.034ms
 2  *  10.150.0.254  754.427ms  *
 3  10.150.0.2  133.038ms  350.054ms  404.197ms
 4  103.170.17.45  312.977ms  403.389ms  311.862ms
 5  180.149.95.33  100.236ms  162.555ms  141.865ms
 6  180.149.92.17  177.978ms  231.933ms  311.882ms
 7  180.149.95.158  403.866ms  204.834ms  312.539ms
 8  38.88.197.184  201.968ms  412.049ms  404.117ms
 9  154.54.25.149  415.594ms  264.541ms  451.323ms
10  154.54.45.161  249.116ms  230.209ms  287.684ms
11  154.54.166.69  455.280ms  313.459ms  403.870ms
12  154.54.165.29  409.326ms  312.541ms  403.979ms
13  154.54.41.54  717.150ms  279.828ms  339.021ms
14  154.54.29.133  512.104ms  404.156ms  312.965ms
15  154.54.169.177  403.706ms  409.792ms  311.950ms
16  154.54.30.122  404.072ms  312.740ms  557.401ms
17  154.54.40.153  363.778ms  278.927ms  335.855ms
18  154.54.170.2  511.669ms  493.422ms  729.137ms
19  154.54.160.70  512.165ms  353.849ms  451.890ms
20  38.32.33.83  444.358ms  598.611ms  403.712ms
21  69.16.0.9  306.748ms  307.585ms  517.090ms
22  69.16.3.0  608.406ms  313.279ms  711.053ms
23  192.80.83.101  415.262ms  511.480ms  404.320ms
24  128.119.0.216  412.365ms  411.547ms  286.679ms
25  128.119.77.233  731.460ms  313.468ms  306.984ms
```

traceroute gaia.cs.umass.edu, cisco.com domain route

Wireshark тусламжтайгаар GET Method capture хийсэн пакет







PC-ээс сэрвэрийн IP хаяг руу нает илгээгээд бүгд received болсон байдал

Сорих асуулт

Өөрийн төхөөрөмжийн IP хаягийг хоёртын тооллын системд хөрвүүлнэ үү. Өөрийн төхөөрөмжийн IP хаягийг 16-тын тооллын системд хөрвүүлнэ үү.

- IP: 10.3.200.241 == 00001010.00000011.11001000.11110001
- IP: 10.3.200.241 == 0A.03.C8.F1

Дүгнэлт

Энэхүү лабораторийн ажлаар би өөрийн төхөөрөмжийн IP хаягийг шинжилж, хөрвүүлэн, сүлжээний дамжууллын орчин, топологи, traceroute үр дүнг судлав.